

Mahmoud Abdellah Salim

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Summary

Software Engineer and Computer Science undergraduate with a strong foundation in backend development, architecting scalable RESTful APIs, microservices, and real-time systems using Node.js, NestJS, TypeScript, and SQL/NoSQL databases. Distinguished by the ability to bridge complex server-side logic with high-performance React.js frontends. Backed by rigorous expertise in data structures, algorithms, and competitive programming (ICPC, Codeforces), I am passionate about solving complex architectural problems and engineering fault-tolerant, highly efficient software solutions.

Education

Suez Canal University
Bachelor of Computer Science and Informatics

Ismailia, Egypt
Sep 2022 – Expected Jun 2026

Projects

Minify Analytics Platform | *NestJS, React, TypeScript, Redux, MongoDB*

API: github.com/Mahmoud142/minify-api | **Client:** github.com/Mahmoud142/minify-web | **Live:** minifyapp.vercel.app

- **Full-Stack Orchestration:** Architected a high-throughput link management platform, bridging a strictly typed, modular NestJS backend with a state-driven React 19 frontend for seamless data synchronization.
- **Telemetry & Data Modeling:** Engineered scalable NoSQL schemas in MongoDB to process concurrent URL resolutions, integrating geolocation pipelines (`geoip-lite`) for granular, real-time analytics.
- **Enterprise Security & Identity Management:** Fortified API infrastructure using strict JWT guards, DTO validation, and distributed rate-limiting. Engineered secure Two-Factor Authentication (2FA) flows via third-party messaging integrations to guarantee rigorous identity verification.

MindMate Healthcare Platform (Graduation Project) | *React, Redux, Socket.io, Leaflet*

Client: [github.com/MindMate-Project/Web](https://github.com/Mahmoud142/MindMate-Project/Web) | **Live:** web-mindmate.vercel.app

- **Real-Time IoT Telemetry:** Spearheaded a low-latency tracking system for Alzheimer's patients, leveraging `Socket.io` to stream live GPS sensor data into a highly responsive Leaflet geospatial mapping interface.
- **Complex State Management:** Designed a fault-tolerant React frontend architecture, utilizing Redux Toolkit to manage asynchronous API integrations and ensure zero-lag synchronization across critical medical dashboards.
- **Secure Access Control:** Implemented rigorous protected routing (React Router v7) and authenticated flows to safeguard sensitive patient telemetry and medication scheduling protocols.

Shopply | *React, Redux Toolkit, Node.js, Express, MongoDB*

API: github.com/Mahmoud142/ECommerce-api | **Client:** github.com/Mahmoud142/Shopply | **Live:** shopplywebapp.vercel.app

- **Full-Stack Architecture:** Engineered a complete e-commerce solution, integrating a responsive React and Redux Toolkit frontend with a production-ready RESTful API to optimize end-to-end data flows and complex cart operations.
- **Relational Design in NoSQL:** Implemented rigorous Role-Based Access Control (RBAC) and architected highly relational MongoDB schemas via Mongoose, ensuring transactional integrity across multi-tier user hierarchies.
- **Third-Party Integration & Optimization:** Integrated the Stripe API for secure financial transactions, Cloudinary for optimized image processing, and SendGrid for automated SMS services.

Skills

Languages: TypeScript, JavaScript (ES6+), Python, C/C++, Java, HTML, CSS

Technology Stack: Node.js, NestJS, Express.js, React.js, Redux Toolkit

Technical Skills: Problem-Solving, Competitive Programming, Data Structure, Algorithms, OOP, Microservices

Soft Skills: Communication, Collaboration, Adaptability, Leadership, Time management, Flexibility, Decision-Making

Databases, DevOps & Tools: SQL, NoSQL, AWS, Docker, Git/GitHub/Gitlab, Linux, Bash, Shell, Scripts, Regex, Postman, Markdown, LaTeX, Microsoft Office, Google Workspace

Achievements & Certifications

- Egyptian Collegiate Programming Contest (ECPC)** | *Contestant*2024 – 2025
- Competed in 2024 and 2025, leveraging teamwork, data structures, and algorithms to solve complex problems under strict time limits.
 - **Verification:** [View Official Contest Standings and Results](#)

- Professional Certifications** | *Udemy, Manara, Index Academy*
NestJS, React, TypeScript & Web Development
- **Verification:** [View All Verified Certificates and Credentials](#)

Languages

- **Arabic:** Native
- **English:** Advanced